1. Physics of Sound

Introduction

This chapter discusses the definition of sound, sound waves, sound power, sound pressure, and sound intensity. The concepts presented in this chapter are the basis for solving industrial noise control problems.

What is Sound?

What is sound? What is noise? Are the two different? Yes, sound is different from noise. Noise is frequently defined as any unwanted sound. The question of whether sound is wanted or not depends not only on the individual but also on the circumstances. What is considered noise (unwanted sound) is therefore subjective. The definition of noise is a functional definition, not a physical definition.

What is sound? Very simply, sound is something that can be heard. Machines make sounds, pianos make sounds, people make sounds, and tuning forks make sounds. Sound is nothing more than an oscillation in atmospheric pressure (a traveling vibratory movement of molecules) within an elastic medium (material) of any phase (gas, liquid or solid). This is a complex definition, containing many, perhaps unfamiliar, terms.

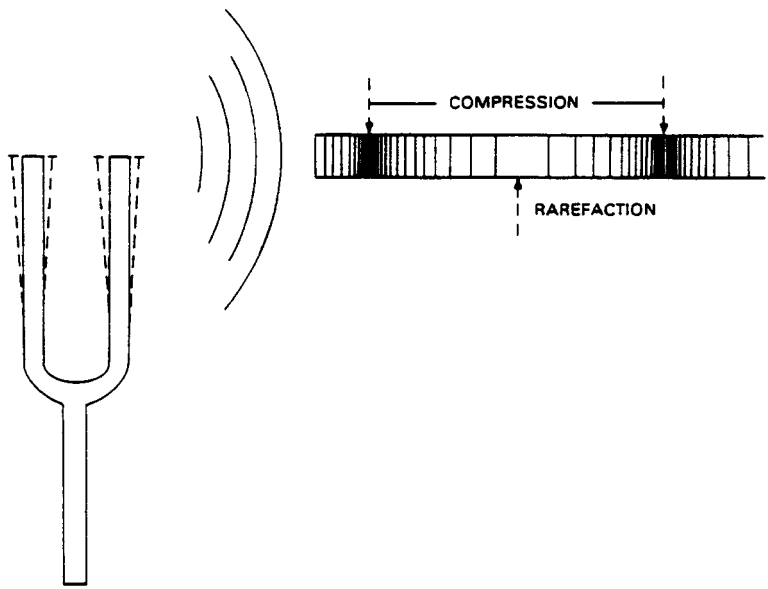
An Example of Sound

Suppose a vibratory machine surface is in an air medium. As the surface vibrates, a sound is heard. What is happening that causes the sound, and why can a sound be heard? As the surface of the machine vibrates in one direction acting as a sound source, the air molecules adjacent to the machine become compressed. This area of compression is an area of increased atmospheric pressure. This area of high pressure acts to push immediately surrounding air molecules at lower pressure. As the surface moves in the opposite direction, the air molecules adjacent to the surface create a low pressure area, an area of rarefaction. So as the machine surface vibrates, it creates alternating areas of compression and rarefaction of the adjacent air molecules. These molecules in turn affect molecules that are adjacent to them. These alternating areas of compression and rarefaction are nothing more than fluctuations in atmospheric pressure. This disturbance in air molecules travels through the air and reaches the listener's ears. As the disturbance reaches the ears, components of the ears begin to vibrate. This mechanical vibration in the ears is eventually changed to electrical energy and is transmitted to the brain, and thus the listener hears the sound.

The areas of compression and rarefaction are illustrated in Figure 4.1.1, with the sound source being a tuning fork.

Figure 4.1.1

Areas of compression and rarefaction.



Sound Wave

As the areas of compression and rarefaction travel in a medium, a sound wave is produced. A sound wave (in an air medium) is the pattern of fluctuation (alternating areas of compression and rarefaction) in the air pressure over a specific distance or time. That is, a sound wave is the pattern of changes in air pressure. This pattern can be graphed as shown in Figure 4.1.2.

In Figure 4.1.2, the vertical axis represents air pressure. The horizontal axis represents either distance or time. By combining Figure 4.1.1 and Figure 4.1.2, it is obvious that there is a correspondence between the wave and the areas of compression and rarefaction of the movement of air molecules. This resulting figure would be as shown in Figure 4.1.3.

The air molecules do not move along the path of the sound wave; rather, the air molecules stay in one place and oscillate around their central location, just as the vibrating source does.

Figure 4.1.2

Pattern of sound--a sound wave.

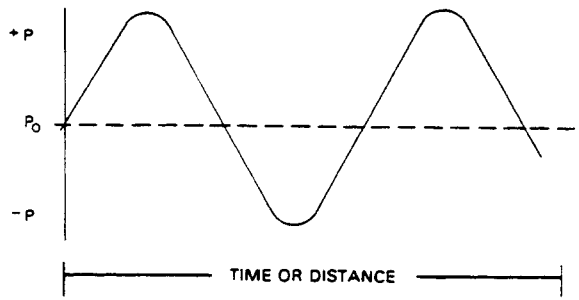
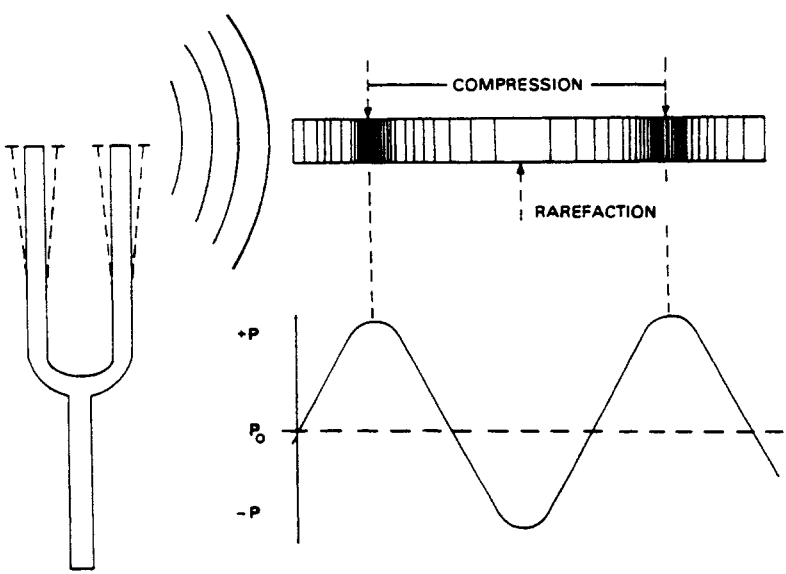


Figure 4.1.3

Areas of compression and rarefaction and a sound wave.



Velocity of Sound

The speed or velocity at which sound travels through a medium is dependent upon (a) the density of the medium (density equals mass divided by volume), and (b) the elasticity of the medium (i.e., the ability of a material to change size and/or shape when subjected to an external force). Mathematically, the relationship between density, elasticity, and velocity is given as

$$C^2 = \frac{\text{elasticity}}{\text{density}}$$

where C denotes velocity.

The above expression is the general expression for the velocity of sound in a medium. In a gas (such as air), the expression can be written as

$$C^2 = \frac{\gamma P_o}{\rho}$$

where C denotes velocity, γ denotes specific heat ratio, ρ denotes density (mass divided by volume), and P denotes atmospheric pressure. Under normal conditions, air at 68°F and $\gamma = 1.41$, it can be shown that the velocity of sound in air is equal to 1127 feet per second or 344 meters per second.

The velocity of sound will be affected by the temperature of the medium. In air the velocity of sound will change about one foot per second for each increase in one degree Fahrenheit (or about two feet per second for each increase in one degree Celsius).

Frequency of Sound

Another important characteristic in describing sound in addition to its velocity is frequency. Frequency refers to the rate at which complete cycles of high and low pressure regions are produced by the sound source. Frequency is usually measured in cycles per second or Hertz (Hz) and is the number of times maximum sound pressure occurs in one second. The ear can hear frequencies between 20 and 20,000 cycles per second.

Period (T) refers to the time it takes for one complete cycle or vibration. It is equal to the reciprocal of frequency.

$$T = \frac{1}{f}$$

where T denotes period and f denotes frequency.

Wavelength

Another interesting characteristic of a sound wave is wavelength. At a given velocity (C), how far would a wave travel in one period (T)? To answer this recall from basic physics that

$$\text{velocity} = \frac{\text{distance}}{\text{time}}$$

where distance refers to how far a wave can travel, and time is represented by the period; that is, the time required for one cycle of pressure change.

Substituting these in the equation gives

$$C = \frac{\lambda}{T}$$

where λ denotes the wavelength and T denotes the period.

Substituting 1/f for the period and solving for wavelength results in the following equation:

$$\lambda = \frac{C}{f}$$

Using the above equation, what is the wavelength of a 1000 Hz wave being propagated through air at 68°F? Recall that the velocity of air under normal conditions is 1127 feet per second.

The answer is as follows:

$$\lambda = \frac{1127 \text{ ft/sec}}{1000 \text{ cycles/sec}}$$

$$\lambda = 1.127 \text{ ft/cycle}$$

This represents the distance a wave would travel in one period (one cycle).

What is the wavelength of a 1000 Hz wave in air at 90°F? Recall that velocity increased one foot per second for each degree Fahrenheit. Therefore, the velocity of sound at 90°F would be 1149 ft/sec.

The answer is as follows

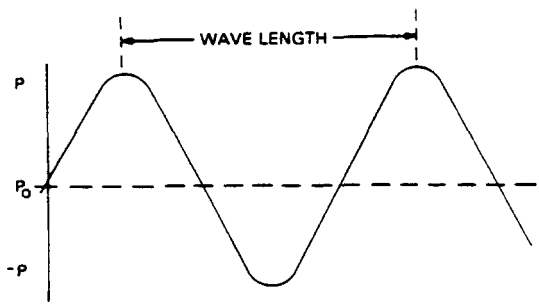
$$\lambda = \frac{1149 \text{ ft/sec}}{1000 \text{ cycles/sec}}$$

$$= 1.149 \text{ ft/cycle}$$

A wavelength can be graphically represented as shown in Figure 4.1.4.

Figure 4.1.4

Representation of a wavelength.



One can further visualize a wavelength by using a spring or a slinky toy. Stretch the spring vertically and then suddenly push down on the top coil and continue to repeat this motion. You will see groupings of compressed coils traveling down the spring. Between these compressed coil groupings will be groupings whose coils are expanded farther apart than usual. Of course, the coils do not move down; they stay in one average position and oscillate about it--the same as the air molecules. The groupings of spring elements move in the wave motion. This motion of the groupings of elements travels with a definite speed. If you change the rate of the up-and-down motion of your hand holding the top of the spring, you will change the frequency of groupings; but the speed of the groupings will stay the same; i.e., the grouping will still travel through the spring at the same speed (the sounds of all frequencies travel at the same speed in the same media). By changing the frequency, you will change the spacing between the groupings. This spacing is the wavelength. The wave speed in the spring remains constant while the wavelength changes. Increasing the frequency decreases the wavelength--the distance between the groupings. Decreasing the frequency increases the wavelength. The same situation occurs with sound in air.

The wave motion in air involves particles (molecules) that vibrate because of the sound energy present. The number of complete cycles per second executed by these molecules is called the frequency of the wave. In the spring analogy, the frequency could be changed, but the speed of sound would not change. This is because speed in a given medium remains constant.

Summary

So far, sound has been defined as an oscillation in pressure within an elastic medium of any phase that evokes an auditory sensation. It has been said that the velocity of sound is dependent upon the density and elasticity

of the medium, where the velocity of sound in normal air is 1127 ft per second. The temperature of the air will affect the velocity of sound (an increase in velocity of one foot per second for each increase in degrees Fahrenheit). Frequency has been discussed and defined as the rate at which complete cycles of high and low pressure regions are produced by the sound source. The period has been defined as the time required for one cycle of pressure change. Thus, the period is equal to the reciprocal of frequency. Wavelength is the distance a wave will travel in one period (one cycle).

It is now time to discuss three important terms that are the basics for understanding the physics of sound: sound pressure, sound power, and sound intensity. Equations to represent these characteristics of sound will be developed, and the relationship between these expressions will be discussed. A conceptual understanding of these terms is important.

Sound Pressure

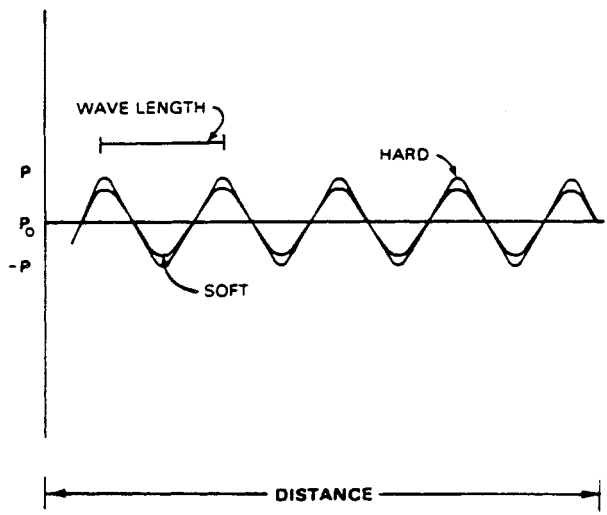
In the previous discussion, sound was defined as an oscillation in atmospheric pressure within an elastic medium of any phase. Sound pressure is defined as the difference between atmospheric pressure and the actual pressure during rarefaction and compression. One can see that sound pressure is unrelated to frequency or wavelength. Sounds in the same medium and with the same frequency may generate different sound pressure. Consider a tuning fork. The fork may be hit lightly and produce a faint sound or may be hit firmly and produce a loud sound. Both sounds will have exactly the same frequency and wavelength since the tines (the forks) will vibrate back and forth the same number of times in a second. The harder the fork is struck, however, the greater the distance the tine travels during each cycle. This increased distance traveled by the tines will cause greater pressure fluctuations above and below atmospheric pressure. This situation is graphically illustrated by Figure 4.1.5.

Figure 4.1.5 illustrates that the harder the tines are struck, the greater the height of the sound wave above atmospheric pressure. It should be noted that the wavelengths are exactly the same. There is a difference between the wave produced by lightly striking the tines and the wave produced by striking the tines a little harder. This difference is the difference in sound pressure. The harder the tines are struck, the greater the fluctuations in atmospheric pressure.

Atmospheric pressure--force per unit area--is usually measured in bars (14.7 pounds per square inch). However, a fluctuation of only 0.1% of atmospheric pressure represents an intolerably loud sound. Since fluctuations in pressure that generate sound are extremely small compared to a bar, it is convenient to use a much smaller unit, the microbar. This unit is abbreviated "μbar" and is equal to one-millionth of a bar. (μbar equals atmospheric pressure times 10^{-6}.) Sound pressure can also be expressed in dynes per square centimeter or newtons per square meter. (One dyne equals 10^{-5} newtons). Normal pressure exerted by the atmosphere is approximately 1,000,000 dynes per square centimeter.

Figure 4.1.5

Wave having same wavelength
but diffferent sound pressure.



The faintest sound that can be detected by the ear is produced by pressure variations of approximately 0.0002 dynes/cm^2 (or 0.0002 newtons/m^2). Discomfort occurs around 20 newtons/m^2, and the threshold of pain is around 200 newtons/m^2. The sounds heard in offices are around one dyne per square centimeter.

The sound pressure generated by a sound source is dependent upon the location of the hearer. The farther away the hearer is from the sound source, the smaller the variations in sound pressure from atmospheric pressure. Because of this, sound pressure is not a good way to evaluate the sound generated by the sound source.

Sound pressure changes over time and is not constant. Since sound pressure is not constant, it is difficult to come up with a single number to describe sound pressure. What has been done is to integrate sound pressure over time. However, most measuring devices do not do this; they use a simpler method, the root mean square (RMS) method. This method amounts to measuring sound pressure at certain places, squaring the sound pressures at those places, summing the sound pressures squared, and dividing by the number of measures taken, and then taking the square root of the average. This method works for pure tone or steady state noise. The time interval over which a single periodic sound pressure pattern must be measured is equal to an integral number of periods of that sound pattern, or the interval must be long compared to a period. There must be more than ten peaks per second for noise to be considered steady state for measurement purposes.

There are two types of measuring instruments: instruments to measure sound pressure in RMS and instruments to measure peak pressures. Maximum peak pressure is normally used when peak pressure is repeated no more than two per second. Peak pressure instruments are used when the noise or sound is not steady state noise or sound.

Sound Power

Sound was described as the movement of air molecules adjacent to a vibratory surface. A tuning fork is the simplest sound generating unit. When it is struck, it begins to vibrate at a definite frequency. Its vibrations transmit energy (where energy is defined as force times distance) to the atmosphere (creating waves having alternating points of maximum and minimum pressure). The amount of energy in the sound is a function of the amount of energy imparted to the tuning fork when it is struck. Every sound source, such as a tuning fork, has a sound power it generates. From basic physics recall that

$$\text{Power} = \frac{\text{energy}}{\text{time}}$$

Sound power can best be defined, then, as the total sound energy radiated by a sound source per unit time.

Energy is usually measured in foot-pounds. A weight of one pound lifted one foot off the ground requires the expenditure of one foot-pound of energy. If it took one second to lift the one pound weight, the power involved would be one foot-pound per second. Power can also be expressed in horsepower, where 550 foot-pounds per second equals one horsepower. In the metric system, power is measured in joules per second, where one joule per second is equal to one watt. A watt is equivalent to about three-quarters of a foot-pound per second.

Table 4.1.1. shows the sound powers radiated by familiar sound sources in watts. (Note: A kilowatt is 1000 watts; a microwatt is one-millionth of a watt.)

Power is also expressed in dyne-centimeters (or newton-meters) per second. One watt equals one joule per second, which equals 10^7 dyne-centimeters per second (one joule equals 10,000,000 dyne-centimeters), or one joule equals one newton-meter (i.e., one joule of energy is expended when one newton of force is exerted over one meter; if it takes one second, then the power involved is one newton-meter per second).

The sound power generated by a machine (sound source) is not measured but calculated from measurements made under controlled laboratory conditions. Manufacturers usually provide the sound power created or generated by their equipment. The industrial hygiene engineer is not responsible for calculating sound power.

Table 4.1.1

Approximate power radiated by some familiar sound sources.

Watts	Prefix Notation	Exponential Notation	Sound Source
10,000,000.	10 megawatts	10^7	
1,000,000.	1 megawatt	10^6	
100,000.	100 kilowatts	10^5	Future turbo-jet (est.)
10,000.	10 kilowatts	10^4	Turbo-jet engine, after-burner
			Turbo-jet engine, take-off
1,000.	1 kilowatt	10^3	2-engine airliner, take-off
100.	100 watts	10^2	Large wood planer
			2-engine airliner, cruising
10.	10 watts	10^1	Chipping hammer, large castings
1.	1 watt	10^0	Chipping hammer, small castings
0.1	100 milliwatts	10^{-1}	Circular saw, ripping plank
0.01	10 milliwatts	10^{-2}	Voice, shouting
0.001	1 milliwatt	10^{-3}	Auto on highway
0.000,1	100 microwatts	10^{-4}	Typewriter
0.000,01	10 microwatts	10^{-5}	Voice, conversational level
0.000,001	1 microwatt	10^{-6}	
0.000,000,1	100 millimicrowatts	10^{-7}	Telephone dial
0.000,000,01	10 millimicrowatts	10^{-8}	Voice whisper
0.000,000,001	1 millimicrowatt	10^{-9}	Pencil writing on paper
0.000,000,000,1	100 micromicrowatts	10^{-10}	
0.000,000,000,01	10 micromicrowatts	10^{-11}	Normal breathing
0.000,000,000,001	1 micromicrowatt	10^{-12}	Man's wrist watch
0.000,000,000,000,1	0.1 micromicrowatt	10^{-13}	

Reprinted from Industrial Hygiene and Toxicology, 1958, Vol. I, 2nd ed., Patty, F.A. (ed.), by permission of John Wiley & Sons, Inc., copyright © 1958.

Remember that sound power is the total sound energy radiated per unit time by the sound source. It is the energy per unit time radiated by the sound source that causes the fluctuations in atmospheric pressure. Therefore, sound power rather than sound pressure is usually used to describe the sound source. However, it should be noted that conceptually sound pressure and sound power are related. At a given distance from a sound source, the increases in sound power will result in increases in sound pressure.

Sound Intensity

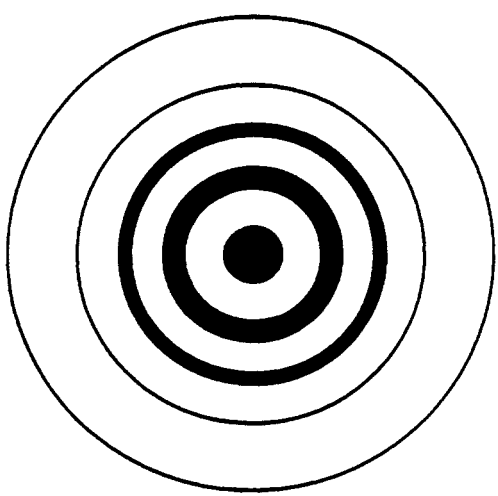
Sound power per unit area is called the sound intensity. If sound power is distributed over an area in such a way that one watt of power falls on each square foot of area, the sound intensity is one watt per square foot. Sound power is an indication of the concentration of power. Thus, when a certain sound power is distributed over a large area, the sound intensity will be much smaller than when the same sound power is concentrated in a smaller area.

Sound intensity is affected by the observer's distance away from the sound source. If a sound source is a point source and that point source radiates a given sound power, the definition of sound intensity makes it clear that the farther away an observer is from that point source, the larger the area in which the sound power is concentrated. Thus, the intensity would become smaller as the observer travels farther from the point source.

When pebbles are dropped into a still pond at a constant rate, the crests of the waves radiated outward are in ever-increasing circles. If it were possible to drop a series of pebbles with proper timing, a pattern similar to that shown in Figure 4.1.6 would be observed. Here the width of the black

Figure 4.1.6

Conceptualization of sound intensity.



rings is intended to indicate the height of the wave. Note that the width of the rings decreases as the diameter increases. Naturally, this is a picture of the surface of the body at only one instant. Since the waves are traveling outward, each ring will expand rapidly until it takes the place of the next larger ring shown in the diagram. This process will continue over and over again so that the picture will always appear to be changing, yet is always the same.

Sound waves radiating from a tuning fork or any other sound source can also be represented by the rings shown in the diagram. Of course, since the sound wave travels in all directions, there will be a pattern similar to this in a vertical plane or in any other plane chosen. Therefore, instead of a set of expanding circular rings there is a set of expanding spherical shells.

In the case of the sound wave, the thickness of the shells corresponds to the intensity of the sound, the concentration of sound power. And, as can be seen, the intensity becomes less as the observer gets farther away from the sound source; i.e., as the diameter of the rings increases, the intensity decreases.

If sound intensity is defined as the sound power per unit area, then the relationship between sound intensity and sound power can be expressed as

$$\text{Sound intensity} = \frac{\text{sound power}}{4\pi r^2}$$

where $4\pi r^2$ represents the surface area of a sphere (r denotes the radius).

If the sound power of a machine is given as 12.5664 foot-pounds per second and an observer is 1.0 feet from the sound source, then the sound intensity at that distance can be computed as

$$\text{Sound intensity} = \frac{12.5664 \text{ ft-pounds/sec}}{4(3.1416)(1.00\text{ft})^2}$$

$$= 1 \text{ foot-pound/sec-ft}^2$$

This means that a sound source of 12.5664 foot-pounds per second will produce a sound intensity of 1 foot pound per square foot at the surface of a hypothetical sphere 1.0 feet from the sound source. This relationship between sound intensity and sound power only holds true in a free-field condition, a condition where the sound is permitted to radiate without bouncing off walls, ceilings, or floors.

The relationship between sound power and sound intensity is often referred to as the inverse square law. From the above expression, it can be noted that if sound power remains constant, the intensity must vary inversely with the square of r, the distance from the source (since π and 4 are always constant).

Assume a sound source that radiates a sound power of 1 watt (one millionth of a watt or one millionth of a joule per second). Further, assume

that the hearer moves away from this sound source a distance of 1 centimeter, then 2 centimeters, then 4 centimeters. Notice that the distances are doubled. Using the formula above, the sound intensity at each of these distances can be computed. At 1 centimeter, the sound intensity would be 0.08 watts per square centimeter; at 2 centimeters, the sound intensity would be 0.02 watts per square centimeter, and at 4 centimeters, the sound intensity would be 0.005 watts per square centimeter. Note that as the distance doubles from the source, the intensity drops by one quarter, or the sound intensity is equal to the inverse of the square of the distance. This relationship is extremely important. It will be used when the solution to noise control problems is discussed.

Sound intensity can also be expressed in terms of sound pressure. Sound intensity is defined as

$$\text{Sound intensity} = \frac{\text{sound power}}{\text{area}} \qquad (4.1.1)$$

Recall that sound power can be expressed as energy per unit time.

$$\text{Sound power} = \frac{\text{energy}}{\text{time}} \qquad (4.1.2)$$

Substituting expression (4.1.2) in expression (4.2.1) above results in

$$\text{Sound intensity} = \frac{\text{energy}}{\text{area x time}} \qquad (4.1.3)$$

From basic physics, recall that

$$\text{Energy} = \text{force x distance} \qquad (4.1.4)$$

Substituting expression (4.1.4) in expression (4.1.3) results in

$$\text{Sound intensity} = \frac{\text{force x distance to move particle}}{\text{area x time to move particle}} \qquad (4.1.5)$$

Recalling from basic physics that

$$\text{Pressure} = \frac{\text{force}}{\text{area}} \qquad (4.1.6)$$

and

$$\text{Velocity} = \frac{\text{distance to move particle}}{\text{time to move particle}} \qquad (4.1.7)$$

Then expression 4.1.5 can be written as

$$\text{Sound intensity} = \text{sound pressure x velocity of the particle} \qquad (4.1.8)$$

The particle velocity in the material of propagation is found from basic impulsive momentum changes.

$$\text{Impulsive force} = \text{change in momentum} \qquad (4.1.9)$$

and

$$\text{Force} \times \text{time} = \text{mass} \times \text{change in particle velocity} \qquad (4.1.10)$$

where

$$\text{Mass} = \text{density} \times \text{volume} \qquad (4.1.11)$$

Substituting (4.1.11) in (4.1.10) results in

$$\text{Force} \times \text{time} = \text{density} \times \text{volume} \times \text{change in particle velocity} \qquad (4.1.12)$$

Rearranging (4.1.12) gives

$$\text{Velocity of particle} = \frac{\text{force} \times \text{time}}{\text{density} \times \text{volume}} \qquad (4.1.13)$$

$$= \frac{\text{force} \times \text{time}}{\text{density} \times \text{area} \times \text{distance}}$$

$$= \frac{\text{pressure}}{\text{density} \times \text{velocity in medium}}$$

Substituting this in expression (4.1.8), results in

$$\text{Sound intensity} = \frac{\text{sound pressure} \times \text{sound pressure}}{\text{density} \times \text{velocity in medium}}$$

or

$$I = \frac{P^2}{\rho C} \qquad (4.1.14)$$

where I denotes sound intensity, P denotes pressure, ρ denotes density of the medium, and C denotes velocity of sound in a medium.

Expression (4.1.14) denotes sound intensity at an instant in time as the impulsive force acts. The total mean sound intensity over time is the integration of

$$I = (1/T \int_0^T P^2 \times dt)/ \rho C$$

By definition, the integral is equivalent to the square of the root mean square (RMS), or

$$I = \frac{P^2_{RMS}}{\rho C}$$

Under normal atmospheric conditions, where I is in μwatts per square feet and P is in μbars, Cox (p. 626, Industrial Hygiene and Toxicology, Volume I) reports the relationship to be

$$I = (1.52P)^2$$

Sound intensity is not steady but varies in a pulsating fashion. With each vibration of the sound source, a pulse of energy is carried from the sound source. One way to measure intensity is to look at the maximum intensity. For a pure tone--such as that resulting from a tuning fork--maximum intensity will be a constant at each compression. For more complicated sounds, this will not be true. The average intensity of a pure sound is equal to one-half the maximum intensity. For more complicated sounds, the average intensity is more meaningful than the maximum intensity. Sound intensity can be averaged. This information will be used later.

Relationship Between Sound Power and Sound Pressure

In the last section, it was shown how sound intensity is related to sound power (by definition).

$$I = \frac{W}{4\pi r^2} \qquad (4.1.1)$$

where I denotes sound intensity, W denotes sound power, r denotes radius of a hypothetical sphere around the sound source. Note: r must be measured in the system being used (either in cm or feet).

In addition, it was shown how sound intensity is related to sound pressure.

$$I = \frac{p2}{C} \qquad (4.1.2)$$

where I denotes sound intensity, P denotes sound pressure (in μbars), denotes density of the medium, and C denotes velocity of sound in the medium.

It is often convenient to express a relationship between sound power and sound pressure. To arrive at this relationship, substitute expression (4.1.2) in expression (4.1.1) and solve for P (sound pressure).

$$\frac{p^2}{\rho C} = \frac{W}{4\pi r^2}$$

$$p^2 = \frac{W\rho C}{4\pi r^2}$$

therefore

$$P = [W\rho C/4\pi r^2]^{1/2} \qquad (4.1.3)$$

If W is given in watts, r in feet, and P in newtons per square meters with standard conditions, equation (4.1.3) can be written as (see The Industrial Environment: Its Evaluation and Control, p. 302).

$$P = [3.5W \times 10^2/r^2]^{1/2} \qquad (4.1.4)$$

The relationship between sound pressure and sound power is very important. It will be used in subsequent chapters.

Given the expression above, predict the sound pressure that would be produced at a distance of 100 feet from a pneumatic chipping hammer. The manufacturer states that the hammer has an acoustic power of 1.0 watts.

Solution

$W = 1.00$ watts

$r = 100$ feet

$P = [3.5 \times 1.0 \text{ watt} \times 10^2/(100 \text{ ft})^2]^{1/2}$

$P = 0.187 \text{ N/m}^2$ (newtons per square meter)

Expression (4.1.4) is only appropriate in a free-field condition. A free-field condition means that the sound created by the sound source is free to travel in all directions; for example, a sound source suspended in the middle of a large room has a free field. The sound would radiate towards the walls, floor, and ceiling without meeting any obstacles or bouncing back. Although a free-field condition is not the normal condition in industrial noise problems, it is the basis for understanding the physics of sound.

Summary

For convenience, the important aspects of this chapter are summarized below.

1. Sound is defined as an oscillation in pressure (a traveling vibratory movement of molecules) within an elastic medium (material) of any phase (gas, liquid, or solid) which evokes an auditory sensation.

2. Noise is subjectively defined as unwanted sound.

3. When a vibratory surface moves back and forth, areas of compression and rarefaction of the adjacent air molecules are created.

4. A sound wave is the pattern of fluctuations in air pressure over distance or time.

5. Velocity of sound depends upon the elasticity and density of the medium.

6. Velocity is given by the following equation:

$$C^2 = \frac{\text{elasticity}}{\text{density}}$$

In air (a gas), the expression can be rewritten as

$$C^2 = \frac{\gamma P_o}{\rho}$$

where γ denotes specific heat ratio and ρ denotes density of the medium.

7. Velocity of sound in normal air is 1127 feet per second, or 344 meters per second.

8. Velocity is affected by temperature. Velocity increases 1 foot per second for each increase in degree Fahrenheit.

9. Frequency (f) is the rate at which complete cycles of high and low pressure regions are produced by the sound source. Frequency is measured in cycles per second or Hertz (Hz).

10. Wavelength is the distance traveled by a wave in one period.

11. A period (T) is the time required for one cycle of pressure change: i.e., $T = 1/f$.

12. The relationship between velocity and wavelength and frequency is given as

$$\lambda = \frac{C}{f}$$

13. Sound pressure is defined as the difference between atmospheric pressure and actual pressure during rarefaction.

14. Sound pressure is not related to frequency. A low frequency sound can have a large sound pressure, and a high frequency sound can have a low pressure.

15. Sound pressure (P) is measured in fractions of bars, usually bars, which is one-millionth of bar. Sound pressure can also be expressed in dynes per square centimeter or newtons per square centimeter.

16. Sound pressure can be measured in root mean squares (RMS) or peak pressures.

17. Sound power (W) is the total sound energy radiated per unit time by a sound source. Sound power is not related to the distance the hearer is away from the sound source as sound pressure is. Sound pressure can be expressed in foot-pounds per second, horsepower, or joules per second (watts).

18. Sound intensity (I) is defined as the sound power per unit area. Sound intensity can be expressed as foot-pounds per second per square foot, watts per square foot, or watts per square centimeter.

19. Sound intensity and sound pressure are related by the inverse square law, which is given by the following equation:

$$W = I4\pi r^2$$

20. Sound intensity and sound pressure are related by the following equation under normal atmospheric conditions (see Industrial Hygiene and Toxicology, Volume I, p. 626).

$$I = (1.52P)^2$$

where I is in μwatts per square foot and P is in μbars.

21. Given that sound intensity is related to sound power and sound pressure, it can be shown that there is a relationship between sound power and sound pressure. This relationship is expressed as

$$P = [W\rho C/4\pi r^2]^{1/2}$$

under normal atmospheric conditions where W is given in watts, C in feet per second, r in feet, and ρ in newtons per square meters. The expression can be written as (see The Industrial Environment: Its Evaluation and Control, p. 302).

$$P = [3.5W \times 10^2/r^2]^{1/2}$$

22. Free-field is defined as a situation where a sound is free to radiate in all directions. The equations developed in this chapter assume a free-field.